

☒ Easy (10 Questions)

1.

What does DoS stand for in cybersecurity?

- A) Denial of Safety
- B) Denial of Service ☒**
- C) Department of Security
- D) Detection of Services

2.

Which of the following is a symptom of a Denial of Service attack?

- A) Increased performance
- B) Slow network or service outage ☒**
- C) More storage space
- D) Enhanced security

3.

What is the key difference between DoS and DDoS?

- A) DoS uses more bandwidth
- B) DoS is legal, DDoS is not
- C) DDoS uses multiple machines, DoS uses one ☒**
- D) DDoS is slower

4.

What is the primary goal of a DoS attack?

- A) To steal data
- B) To make a service unavailable ☒**
- C) To increase speed
- D) To update software

5.

Buffer overflows occur when a program:

- A) Receives unexpected input
- B) Is infected by malware
- C) Writes more data than a buffer can hold ☒**
- D) Runs too fast

6.

What is input validation primarily used for?

- A) Debugging code
- B) Preventing invalid or malicious inputs ☒**
- C) Encrypting data
- D) Logging user input

7.

Which of the following is an example of proper input validation?

- A) Accepting any user input
- B) Using eval() to process strings
- C) Rejecting inputs with unexpected characters** ☒
- D) Storing input without checking

8.

Access control is used to:

- A) Encrypt data
- B) Restrict who can access data or resources** ☒
- C) Monitor network packets
- D) Increase performance

9.

Which of the following is a type of access control?

- A) Mandatory
- B) Discretionary
- C) Role-Based
- D) All of the above** ☒

10.

Which of the following is a strong access control mechanism?

- A) Username only
- B) IP restriction only
- C) Multi-factor authentication** ☒
- D) Hardcoded password

☒ Medium (20 Questions)

11.

Which tool is often used to launch a DoS attack?

- A) Wireshark
- B) LOIC (Low Orbit Ion Cannon)** ☒
- C) Nmap
- D) Nessus

12.

How can a DoS attack affect businesses?

- A) Improve uptime
- B) Cause financial loss and reputation damage** ☒
- C) Reduce bandwidth
- D) Help test performance

13.

Which of the following best describes a SYN flood attack?

- A) Encrypts files
- B) Exploits the TCP handshake by sending many SYN packets** ☒
- C) Steals cookies
- D) Alters headers

14.

Buffer overflows can lead to:

- A) Password reset
- B) Code execution or crashes** ☒
- C) Slower CPU
- D) Memory defragmentation

15.

What type of input should be rejected during input validation?

- A) Alphanumeric values
- B) Valid email addresses
- C) Unexpected characters or patterns** ☒
- D) Strong passwords

16.

Which of the following techniques prevents buffer overflow?

- A) Memory swapping
- B) Bounds checking** ☒
- C) Increasing RAM
- D) Using arrays

17.

What is the goal of role-based access control (RBAC)?

- A) Assign access to specific IP addresses
- B) Grant permissions based on user roles** ☒
- C) Hide logs from users
- D) Prevent logins during business hours

18.

Which of the following is a valid form of input validation?

- A) Blacklisting certain inputs
- B) Whitelisting valid inputs only** ☒
- C) Accepting all inputs
- D) Using eval()

19.

What is a potential danger of buffer overflows in C programs?

- A) Stack trace output
- B) Execution of arbitrary shellcode** ☒
- C) Infinite loop
- D) Program termination only

20.

Which programming language is most prone to buffer overflows?

- A) Python
- B) Java
- C) C/C++ ☒
- D) PHP

21.

Access control that grants rights based on user identity is:

- A) MAC
- B) DAC ☒
- C) RBAC
- D) ABAC

22.

Which header can help mitigate DoS attacks from repeated requests?

- A) Content-Type
- B) User-Agent
- C) Retry-After ☒
- D) Authorization

23.

Which tool can detect a buffer overflow vulnerability?

- A) Burp Suite
- B) Valgrind ☒
- C) Nmap
- D) Nikto

24.

What type of attack is characterized by sending oversized input to crash a system?

- A) SQL Injection
- B) XSS
- C) Buffer Overflow ☒
- D) DoS

25.

Which of the following can mitigate buffer overflows?

- A) Debugging
- B) Stack canaries ☒
- C) Caching
- D) Web proxies

26.

In mandatory access control (MAC), who defines access policies?

- A) Users
- B) System administrator or OS ☒

- C) External auditor
- D) Firewalls

27.

Which of these is not an access control method?

- A) RBAC
- B) DAC
- C) ABAC
- D) DNS ☒

28.

Which phase of software development is best for applying input validation?

- A) Deployment
- B) Design and development ☒
- C) Debugging
- D) Testing

29.

What is rate limiting used for?

- A) Encrypting API requests
- B) Throttling requests to prevent abuse (e.g., DoS) ☒
- C) Preventing SQL injection
- D) Tracking users

30.

In access control, the principle of least privilege means:

- A) Giving users admin rights
- B) Granting the minimum level of access required ☒
- C) Removing passwords
- D) Disabling firewalls

☒ Hard (10 Questions)

31.

What is heap spraying commonly used for in buffer overflow attacks?

- A) Logging memory usage
- B) Placing shellcode in predictable locations in memory ☒
- C) Clearing RAM
- D) Encrypting buffers

32.

Which of the following mitigates DoS attacks at the network level?

- A) Input validation
- B) Firewalls and rate limiting** ☒
- C) HTML sanitization
- D) TLS

33.

What is ASLR and how does it help?

- A) A protocol for rate limiting
- B) Randomizes memory addresses to hinder buffer overflows** ☒
- C) Validates access rights
- D) Encodes URLs

34.

A buffer overflow in which memory region allows code execution most easily?

- A) Heap
- B) Stack
- C) Stack (when executable)** ☒
- D) Data section

35.

What type of access control combines multiple attributes for decisions?

- A) MAC
- B) DAC
- C) ABAC (Attribute-Based Access Control)** ☒
- D) IPAC

36.

What is a smurf attack?

- A) SQL attack
- B) A type of DoS using ICMP floods and IP spoofing** ☒
- C) Ransomware
- D) Access control bypass

37.

Which memory corruption exploit bypasses stack protection?

- A) Format string
- B) Return-Oriented Programming (ROP)** ☒
- C) TLS downgrade
- D) Race condition

38.

How does CAPTCHA help prevent DoS?

- A) Encrypts data
- B) Blocks automated requests** ☒
- C) Monitors traffic
- D) Validates passwords

39.

Which type of access control is most scalable in enterprise settings?

- A) DAC
- B) RBAC** ☒
- C) MAC
- D) Static rules

40.

In a buffer overflow exploit, what does NOP sled refer to?

- A) Stack protector
- B) Debugger trap
- C) Padding leading to shellcode execution** ☒
- D) Memory sanitizer